

MARKET RISK QUANT NOTES • 2026 EDITION

You don't need another VaR definition. You need **the desk.**

VaR, Expected Shortfall, Backtesting & FRTB for market-risk quants**19**

modules + 3-project lab

20

runnable Python labs

64

interview Q&As

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Built for

Market-risk analysts, risk-methodology and FRTB quants, model validators, and anyone interviewing for a risk seat at a bank, a PD or a consultancy.

Every number is reproducible

All worked examples, tables and charts come from the companion files `mrq_lab.py`, `mrq_examples.py` and `mrq_figures.py`.

What's inside

Desk-to-capital coverage

01	The market-risk job	daily cycle, roles, P&L types	5
02	Risk factors, sensitivities & P&L	shifts, Greeks, full vs sensitivity revaluation	11
03	VaR foundations & parametric VaR	quantiles, scaling, component VaR	18
04	Non-linear books	delta-gamma, Cornish-Fisher, full revaluation	26
05	Historical simulation	conventions, ghosts, weighting, FHS	33
06	Monte Carlo VaR	Cholesky, fat tails, convergence	42
07	Volatility & correlation	EWMA, GARCH, correlation breakdown	50
08	Expected Shortfall	coherence, estimation, ES backtests	58
09	Backtesting	traffic light, Kupiec, Christoffersen	66
10	Stressed VaR & stress testing	Basel 2.5, scenario design	76
11	FRTB architecture	boundary, desks, SA vs IMA	81
12	FRTB SA: sensitivities-based method	delta, vega, curvature	86
13	FRTB SA: DRC & RRAO	jump-to-default, residual risk	96
14	FRTB IMA: ES & liquidity horizons	cascade, stress calibration, IMCC	101
15	FRTB IMA: model eligibility	PLA test, desk backtests, RFET, NMRF	107
16	Production, validation & case studies	what breaks in real banks	114
17	Risk reporting & communication	daily report, commentary, breach templates	120
18	Interview drill	64 questions with desk-grade answers	123
19	Problem set	18 worked problems, pen first	129
A-D	Appendices	formula sheet, code map, plan, references, full driver script	132

COMPANION CODE

`mrq_lab.py` holds every function (≈ 400 lines, numpy + scipy). `mrq_examples.py` runs Labs 1–17 and prints every number quoted in the notes. `mrq_figures.py` redraws every chart. Run them in that order.

Concept deep-dives

3.6 The concepts, layer by layer

Concept 3.1 Parametric VaR in one line

PLAIN-ENGLISH INTUITION

If P&L is normal, every quantile is a fixed multiple of its standard deviation. So VaR is ‘how many standard deviations’ times ‘how big is a standard deviation’.

CORE EQUATION

$$\text{VaR}_\alpha = z_\alpha \sigma_P \sqrt{h}, \quad \sigma_P = \sqrt{e^\top \Sigma e}$$

WHAT THE SYMBOLS MEAN

- z_α standard normal quantile
- e rupee exposures to factor returns
- Σ covariance of daily factor returns
- h horizon in days

DIRECT PROPORTIONAL RELATIONSHIPS

exposure e (all scaled by λ)	$\propto$	VaR scales by λ
volatility of each factor	$\uparrow$	VaR $\uparrow$
confidence α	$\uparrow$	$z_\alpha \uparrow$, VaR $\uparrow$ (2.326 at 99% vs 1.645 at 95%: +41%)
horizon h	$\propto$	VaR $\propto \sqrt{h}$ (iid only)
correlation between two long positions	$\uparrow$	VaR $\uparrow$
correlation between a long and a hedge	$\uparrow$	VaR $\downarrow$

Tiny worked example. $\sigma_P = ₹11.51$ lakh. 99% VaR = $2.3263 \times 11.51 = ₹26.78$ lakh; 95% VaR = $1.6449 \times 11.51 = ₹18.93$ lakh; 10-day 99% = $26.78 \times 3.162 = ₹84.67$ lakh.

MEMORY ANCHOR

VaR = Z * Sigma * Root-h — “ZSR”

Three dials: confidence (Z), risk size (Sigma), horizon (Root-h). Change one at a time in interviews.

Concept 3.2 Diversification benefit

PLAIN-ENGLISH INTUITION

Two risks that do not move together partly cancel. The benefit is the gap between adding stand-alone VaRs and computing VaR of the sum.

CORE EQUATION

$$\text{benefit} = \sum_i \text{VaR}_i - \text{VaR}\left(\sum_i X_i\right) = z_\alpha \left(\sum_i \sigma_i - \sigma_P\right) \geq 0$$

WHAT THE SYMBOLS MEAN

σ_i stand-alone rupee volatility of position i

DIRECT PROPORTIONAL RELATIONSHIPS

correlation ρ between long positions	↓	benefit ↑
number of independent positions n (equal size)	↑	$\sigma_P \propto \sqrt{n}$ while $\sum \sigma_i \propto n$
crisis ($\rho \rightarrow 1$)	↓	benefit $\rightarrow 0$

Tiny worked example. Two-asset book: $\sum$ stand-alone = $2.326 \times (12 + 2.4) = ₹33.50$ lakh; VaR = ₹26.78 lakh; benefit ₹6.72 lakh (20%). At $\rho = +1$ the benefit is 0.

MEMORY ANCHOR

Sum of VaRs $\geq$ VaR of Sum (for normals)

“Sum-of $\geq$ Of-sum”. The inequality is the diversification benefit; it fails for VaR with non-elliptical losses (Module 8).

Concept 3.3 Euler allocation: marginal and component VaR

PLAIN-ENGLISH INTUITION

Because VaR scales one-for-one with the whole book, you can split it into pieces, one per position, that add up exactly. A piece is the position's size times how fast VaR grows as you add to it.

CORE EQUATION

$$C_i = e_i \frac{\partial \text{VaR}}{\partial e_i} = z_\alpha \frac{e_i (\Sigma e)_i}{\sigma_P}, \quad \sum_i C_i = \text{VaR}, \quad \frac{C_i}{\text{VaR}} = \frac{\text{Cov}(X_i, X_P)}{\text{Var}(X_P)}$$

WHAT THE SYMBOLS MEAN

C_i component VaR of position i
 $(\Sigma e)_i$ covariance of factor i with the portfolio

DIRECT PROPORTIONAL RELATIONSHIPS

covariance of a position with the book	$\propto$	its component VaR
a hedge has negative covariance, so a negative component		
position's stand-alone vol	↑	component VaR ↑ only if correlated with the book

Tiny worked example. Components ₹27.36 lakh (Nifty) and -₹0.58 lakh (USD forward) = ₹26.78 lakh. The forward is a hedge: cutting it raises VaR.

MODULE 09

Backtesting: the day your model meets P&L

A VaR model is a forecast. Backtesting is how the forecast is graded, by the bank and by the supervisor, with capital at stake.

- ✓ Place a model in the Basel traffic-light zones and compute its capital multiplier
- ✓ Test for clustered exceptions with Christoffersen's independence test
- ✓ Run Kupiec's coverage test by hand, including the zero-exception case
- ✓ Explain why backtests have low power and what banks do about it

BEGINNER FIRST-PRINCIPLES PRIMER

A backtest compares yesterday's VaR with today's P&L. A correct 99% model is breached on about 1 day in 100, and the breaches arrive randomly, not in clusters. Count them (Kupiec), check they do not cluster (Christoffersen) and map the count to a capital multiplier (traffic light).

HOW TO STUDY THIS MODULE IF YOU ARE NEW

1. Memorise the zones: green 0–4, yellow 5–9, red 10+.
2. Compute Kupiec for 8 exceptions by hand.
3. Read the lab's 20-year backtest table and say which model you would approve.

DESK2QUANT STANDARD

For every idea in this module, connect five layers out loud:

hit sequence → iid Bernoulli($1-\alpha$) → VaR at t , P&L at $t+1$ → low power, clustering → multiplier and model approval

MODULE MAP

- | | |
|--|--|
| 9.1 Hits, and the two things a good model must get right | 9.5 Christoffersen's independence test |
| 9.2 The Basel traffic light | 9.6 Walk-forward comparison: HS vs EWMA vs FHS |
| 9.3 The power problem | 9.7 Exceptions in practice |
| 9.4 Kupiec's proportion-of-failures test | 9.8 The concepts, layer by layer |

The set-up

9.1 Hits, and the two things a good model must get right

For each day t compare the VaR forecast made the evening before with the next day's P&L, and record a **hit**:

FORMULA

$$I_t = \mathbf{1}\{P\&L_t < -\text{VaR}_{t|t-1}\}, \quad \text{correct model: } I_t \stackrel{iid}{\sim} \text{Bernoulli}(p), \quad p = 1 - \alpha$$

Christoffersen (1998) split "correct" into two testable properties:

Unconditional coverage

The exception rate equals p on average: about 2.5 exceptions per 250 days at 99%. Too many means VaR is too low; too few means it is too high (and capital is wasted).

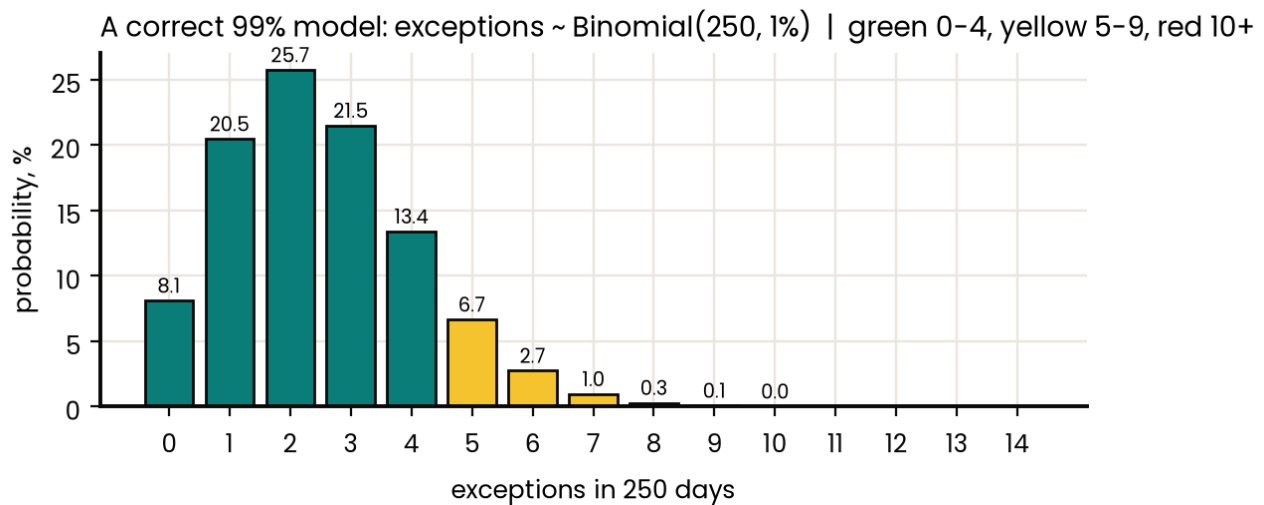
Independence

Exceptions do not cluster. An exception today should not make one tomorrow more likely. Clusters mean VaR reacts too slowly to volatility.

Which P&L? Hypothetical P&L tests the model on a frozen book; actual P&L tests the frozen-book assumption as well (Module 1). Supervisors want both; the timing must match exactly: yesterday's VaR, today's P&L, same portfolio scope.

9.2 The Basel traffic light

With 250 days at 99%, a correct model's exception count is $X \sim \text{Bin}(250, 0.01)$. The Basel Committee's 1996 framework, carried into Basel 2.5, turns the count into a **capital multiplier** on the 60-day average VaR:



Probabilities of each exception count for a correct 99% model over 250 days, coloured by zone.

Zone	Exceptions	Plus factor	Multiplier k	$P(X \leq x)$, correct model
Green	0 - 4	0.00	3.00	up to 89.22%
Yellow	5	0.40	3.40	95.88%
Yellow	6	0.50	3.50	98.63%
Yellow	7	0.65	3.65	99.60%
Yellow	8	0.75	3.75	99.89%
Yellow	9	0.85	3.85	99.97%
Red	10+	1.00	4.00	99.99%

MODULE 12

The sensitivities-based method, **step by step**

SBM turns a desk's Greeks into capital with prescribed risk weights and correlations. Every step is mechanical, which is exactly why interviewers ask you to do it on paper.

- ✓ Define FRTB delta and vega sensitivities for each risk class, with units
- ✓ Explain why the low-correlation scenario binds for a curve spread and the high one for a long-only book
- ✓ Aggregate weighted sensitivities within and across buckets
- ✓ Compute a curvature charge for a short option and show a long option gets none

BEGINNER FIRST-PRINCIPLES PRIMER

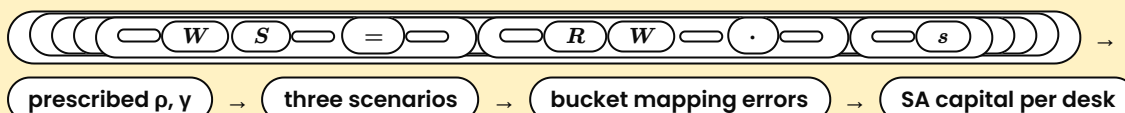
SBM is Greeks times prescribed shocks, combined with prescribed correlations, three times (low, medium, high correlation), keeping the worst total. It is mechanical, which is why interviews ask you to do it on paper.

HOW TO STUDY THIS MODULE IF YOU ARE NEW

1. Convert a PV01 into an FRTB sensitivity and a weighted sensitivity.
2. Compute K_b for a two-tenor curve trade in all three scenarios.
3. Explain why a long option has no curvature charge.

DESK2QUANT STANDARD

For every idea in this module, connect five layers out loud:



MODULE MAP

- | | |
|---|--|
| 12.1 Seven risk classes, three measures | 12.6 FX and equity delta |
| 12.2 Sensitivity definitions | 12.7 Vega |
| 12.3 The aggregation recipe | 12.8 Curvature |
| 12.4 GIRR in detail | 12.9 The concepts, layer by layer |
| 12.5 Worked example: a 5s10s curve trade | |

Check	What it catches	Typical rule
Calendar alignment	Mixing Indian and US holidays in one scenario date	One scenario calendar; defined treatment for partial holidays
Snapshot time	Asia close vs London close for the same factor	One official snapshot per factor; document asynchronicity
Position reconciliation	Trades missing from the risk feed, duplicates, wrong notional sign	Daily count and notional reconciliation vs front office

DESK REALITY

In an HS engine a single bad price does not produce one bad day. It produces a bad **scenario** that is applied to today's book for the next 250 or 500 days. A spike that lands in the tail can set VaR, and set ES, for a year. Scenario-level cleaning, with an audit trail, matters more than any methodology choice.

16.2 Explaining a VaR move

When VaR moves from V_{t-1} to V_t , change one input at a time, in a fixed order, and attribute the difference:

1. **Positions:** yesterday's scenarios, yesterday's market, today's positions. Difference = new trades and maturities.
2. **Market level:** add today's market data (revaluation base). Difference = moves in spot, vol, curves that change sensitivities.
3. **Scenario set:** roll the window (drop the oldest day, add yesterday). Difference = roll-off and roll-in effects.
4. **Methodology / data fixes:** anything left, including corrections, proxy changes and model releases.

The order matters (the attribution is path-dependent), so the bank fixes one order and keeps it. For HS, report the **VaR-setting scenario date** on both days: if it changed, the move is about the tail composition, not the level of risk.

16.3 Validating a VaR/ES model

Model-risk guidance (for example the US SR 11-7 guidance of 2011 and the UK PRA's SS1/23) asks for the same core elements. For a market-risk model they translate into:

Element	Questions a validator answers
Conceptual soundness	Are the method, window, shift conventions, proxies and liquidity horizons appropriate for the portfolio? What does the model ignore?
Data	Are histories complete, clean and representative? How many factors are proxied or filled, and with what basis risk?
Implementation	Does the code do what the documentation says? Independent re-implementation on sample portfolios; pricing parity with front office.
Outcomes analysis	Backtesting at several confidence levels and portfolio levels; PLA; distribution tests beyond exception counts.

MODULE 17

Risk reporting: the number is half the job

A risk number nobody understands by 9 a.m. changes nothing. This module is the desk half of the job: what the daily report contains, how to write commentary, how to escalate a breach and how to present to a risk committee.

- ✓ Lay out a one-page daily market-risk report
- ✓ Draft a limit-breach notification and a backtesting-exception note
- ✓ Write VaR commentary in the What / Why / So what / Now what format
- ✓ Build a weekly committee pack that leads with decisions

The daily report

17.1 Anatomy of a one-page daily risk report

Block	Content	Reader's question
Headline	Firm VaR, ES, change vs yesterday, utilisation; one sentence on the main driver	Is anything wrong?
Desk table	VaR, limit, utilisation, day-on-day change, stressed VaR per desk; breaches in red, > 80% in amber	Where is the risk?
VaR explain	Change split into positions, market, scenario roll, data/model (Module 16)	Why did it move?
Sensitivities	Top PV01 buckets, FX deltas, vega by expiry, largest single-name CS01	What is the desk actually betting on?
Stress	Worst three scenarios for the firm and the desks driving them	What if the window is wrong?
Backtesting	Yesterday's P&L vs VaR, exceptions YTD and last 250 days, zone	Is the model still right?
Data & issues	Stale prices, proxies used, late trades, open model findings	Can I trust today's number?

DESK REALITY

Senior readers spend under a minute on the report. Put the answer in the headline sentence, keep one page, and move detail to appendices. A report that needs explaining in a meeting has already failed.